To see that (5) $\Rightarrow$ (1), note that if A is a completely reducible representation of A, then each element of $\mathrm{Rad}(A)$ kills it, but the only element that kills $1 \in A$ is 0; thus $\mathrm{Rad}(A) = 0$, so A is semisimple. $\square$

3.6. Characters of representations

Let A be an algebra and V a finite dimensional representation of A with action ρ. Then the **character** of V is the linear function $\chi_V : A \to k$ given by

$$\chi_V(a) = \mathrm{Tr}|_V(\rho(a)).$$

If $[A, A]$ is the span of commutators $[x, y] := xy - yx$ over all $x, y \in A$, then $[A, A] \subseteq \ker \chi_V$. Thus, we may view the character as a mapping $\chi_V : A/[A, A] \to k$.

Exercise 3.6.1. Show that if $W \subset V$ are finite dimensional representations of A, then $\chi_V = \chi_W + \chi_{V/W}$.

Theorem 3.6.2. *(i) Characters of (distinct) irreducible finite dimensional representations of A are linearly independent.*

(ii) If A is a finite dimensional semisimple algebra, then these characters form a basis of $(A/[A, A])^*$.

Proof. (i) If $V_1, \ldots, V_r$ are nonisomorphic irreducible finite dimensional representations of A, then the map

$$\rho_{V_1} \oplus \cdots \oplus \rho_{V_r} : A \to \mathrm{End}\ V_1 \oplus \cdots \oplus \mathrm{End}\ V_r$$

is surjective by the density theorem, so $\chi_{V_1}, \ldots, \chi_{V_r}$ are linearly independent. (Indeed, if $\sum \lambda_i \chi_{V_i}(a) = 0$ for all $a \in A$, then $\sum \lambda_i \mathrm{Tr}(M_i) = 0$ for all $M_i \in \mathrm{End}_k V_i$. But each $\mathrm{Tr}(M_i)$ can range independently over k, so it must be that $\lambda_1 = \cdots = \lambda_r = 0$.)

(ii) First we prove that $[\mathrm{Mat}_d(k), \mathrm{Mat}_d(k)] = \mathfrak{sl}_d(k)$, the set of all matrices with trace 0. It is clear that $[\mathrm{Mat}_d(k), \mathrm{Mat}_d(k)] \subseteq \mathfrak{sl}_d(k)$. If we denote by E_{ij} the matrix with 1 in the ith row of the jth column and 0's everywhere else, we have $[E_{ij}, E_{jm}] = E_{im}$ for $i \neq m$ and $[E_{i,i+1}, E_{i+1,i}] = E_{ii} - E_{i+1,i+1}$. Now $\{E_{im}\} \cup \{E_{ii} - E_{i+1,i+1}\}$ forms a basis in $\mathfrak{sl}_d(k)$, so indeed $[\mathrm{Mat}_d(k), \mathrm{Mat}_d(k)] = \mathfrak{sl}_d(k)$, as claimed.

By semisimplicity, we can write $A = \mathrm{Mat}_{d_1}(k) \oplus \cdots \oplus \mathrm{Mat}_{d_r}(k)$. Then $[A, A] = \mathfrak{sl}_{d_1}(k) \oplus \cdots \oplus \mathfrak{sl}_{d_r}(k)$, and $A/[A, A] \cong k^r$. By Theorem